

Electrostatic transition state stabilization rather than reactant destabilization provides the chemical basis for efficient chorismate mutase catalysis

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For more than half a century, transition state theory has provided a useful framework for understanding the origins of enzyme catalysis. As proposed by Pauling, enzymes accelerate chemical reactions by binding transition states tighter than substrates, thereby lowering the activation energy compared with that of the corresponding uncatalyzed process. This paradigm has been challenged for chorismate mutase (CM), a well-characterized metabolic enzyme that catalyzes the rearrangement of chorismate to prephenate. Calculations have predicted the decisive factor in CM catalysis to be ground state destabilization rather than transition state stabilization. Using X-ray crystallography, we show, in contrast, that a sluggish variant of *Bacillus subtilis* CM, in which a cationic active-site arginine was replaced by a neutral citrulline, is a poor catalyst even though it effectively preorganizes chorismate for the reaction. A series of high-resolution molecular snapshots of the reaction coordinate, including the apo enzyme, and complexes with substrate, transition state analog and product, demonstrate that an active site, which is only complementary in shape to a reactive substrate conformer, is insufficient for effective catalysis. Instead, as with other enzymes, electrostatic stabilization of the CM transition state appears to be crucial for achieving high reaction rates.

pericyclic reaction | catalysis | enzyme mechanism | near attack conformation | X-ray crystal structures

Nearly seven decades ago, Pauling proposed that an enzyme preferentially binds and stabilizes the transition state of the chemical reaction it catalyzes relative to the substrate in the ground state (1). As a consequence, the activation energy is lowered for the catalyzed versus the uncatalyzed process, giving rise to large rate accelerations. This simple model has successfully informed the development of potent enzyme inhibitors (2), catalytic antibodies (3), and computationally designed catalysts (4, 5). Nevertheless, due to the complexity of even the simplest enzymatic systems, a quantitative understanding of the energetics of catalysis has remained elusive. The relative contributions of electrostatic, steric, proximity, and solvent effects to transition state stabilization and reactant destabilization, as well as the role of dynamic motions and tunneling, are still heatedly debated (6–10).

Chorismate mutase (CM) has become a popular model system for studies on enzyme catalysis. It accelerates the unimolecular conversion of chorismate (1) to prephenate (3) (Fig. 1A), a key step in the biosynthesis of tyrosine and phenylalanine, 2×10^6 -fold over the spontaneous reaction in water (11). Both the enzymatic and nonenzymatic processes have been shown to be concerted Claisen rearrangements with asynchronous chairlike transition states (2) (12, 13). Because chorismate exists in an extended pseudodiequatorial conformation (1a) in aqueous solution (14), accessing the transition state first requires that the substrate undergo an energetically unfavorable conformational change to populate a rare pseudodiaxial conformer (1b) in which the enolpyruvyl side chain is positioned over the cyclohexadiene ring. The

ability of a protein to populate this reactive ground state conformer has been proposed to account almost entirely for the efficiency of CM catalysis (15–17).

Structural studies have shown that CMs use an extensive array of hydrogen-bonding and electrostatic interactions to bind the pseudodiaxial substrate conformer (18–20). Sterically constraining chorismate in a “near attack conformation” (NAC) (15–17, 21), in which the reacting centers are confined to contact distances, would be expected to increase the probability of reaction. Ground state destabilization through conformational compression has been shown to afford large rate accelerations for Claisen rearrangements in synthetic model systems (22). Based on molecular dynamics (MD), quantum mechanics/molecular mechanics (QM/MM), and thermodynamic integration studies, Bruice and coworkers estimated that selective NAC formation at the enzyme active site accounts for ~90% of the kinetic advantage provided by CM and concluded that transition state binding is unimportant for catalysis in this system (15–17). In contrast, other QM/MM studies (23, 24), including semiempirical and higher level calculations, have suggested that NAC formation is the result of transition state stabilization, not the source of catalysis. Polar active site residues, most notably a cationic arginine or lysine positioned next to the ether oxygen of the

Significance

Chorismate mutase (CM) is a textbook model for enzyme catalysis. Although it promotes a simple unimolecular reaction, the origins of its 2-million-fold rate acceleration have been debated for decades. The relative importance of electrostatic transition state stabilization versus ground state destabilization has been a particularly contentious issue. High-resolution crystallographic snapshots of an engineered CM variant and its complexes with substrate, transition state analog, and product now provide strong experimental evidence that properly positioned active-site charges are essential in this system and that preorganization of the substrate in a reactive conformation contributes relatively little to catalysis. A proper understanding of the role of electrostatics in this and other enzymes is important for ongoing efforts to design new enzymes de novo.

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Data deposition: The apo BsCM crystal structure and the Arg90Cit BsCM* complexes have been deposited in the RCSB Protein Data Bank, www.rcsb.org (PDB ID codes 3ZOP, 3ZP4, 3ZP7, and 3Z08).

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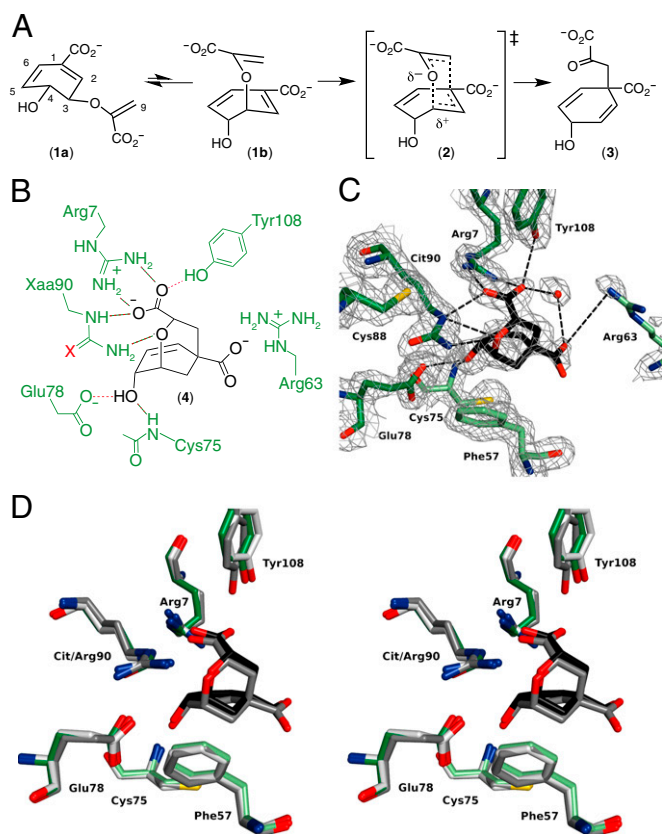


Fig. 1. CM reaction and enzyme active site. (A) The Claisen rearrangement of chorismate (1a and 1b) to prephenate (3) occurs via a chair-like pericyclic transition state (2). (B) Schematic view of the active site of BsCM* (X = NH₂⁺) and Arg90Cit BsCM* (X = O) in complex with the conformationally constrained transition state analog 4. (C) A σ_A -weighted $2F_o - F_c$ map of the Arg90Cit BsCM* active site in complex with 4. The electron density is contoured at 1.5 σ ; for clarity, portions of the enzyme obscuring the view of the ligand are not shown. Residues from different subunits are shown in light and dark green. Dashed lines indicate H-bonding contacts. A water molecule, bridging the carboxylate groups of the ligand, is shown as a red sphere. (D) Stereoview of the Arg90Cit BsCM* complex (protein in green; ligand in black) superimposed on the analogous complex of WT BsCM [PDB ID code 2CHT (18); protein and ligand in gray] and the newly determined apo structure of WT BsCM (white).

breaking C–O bond, were proposed to stabilize the high-energy transition state electrostatically relative to the bound substrate.

Kienhöfer et al. experimentally probed the importance of electrostatic effects in this system using a semisynthetic variant of the CM from *Bacillus subtilis* (BsCM) (25). Seeking to separate ground state from transition state effects, they replaced the positively charged Arg90 residue at the active site of BsCM (Fig. 1B) with a noncharged arginine analog, the amino acid citrulline, which is not normally found in proteins. Consistent with the hypothesis that Arg90 stabilizes a developing partial negative charge in the transition state (18, 26–28), the Arg90Cit substitution has little effect on the ground state Michaelis complex, judging from the mere threefold increase in K_m for chorismate (note that K_m approximates K_s for WT BsCM) (29), but it greatly impairs catalytic efficiency, reducing both k_{cat} and k_{cat}/K_m by more than 4 orders of magnitude (25). These changes suggest that the positively charged arginine contributes up to 5.9 kcal/mol to transition state stabilization but only 0.6 kcal/mol to the binding energy of the ground state. This interpretation has been challenged by Hur and Bruice (17), however. Based on free-energy MD simulations, they have argued that the poor activity

of the Arg90Cit variant is not due to inferior transition state stabilization but rather to major structural rearrangements at the active site that adversely affect the efficiency of forming reactive conformers.

Here we present structural evidence that bolsters the findings from biochemical characterization of the Arg90Cit mutant. High-resolution crystallographic snapshots of BsCM and its ligand complexes provide direct insight into key species along the reaction coordinate. As seen with many other enzymes (8), electrostatic transition state stabilization rather than reactant destabilization appears to be the determining factor in CM catalysis.

Results

To resolve the debate about the relative importance of transition state stabilization for CM catalysis, we set out to determine the crystal structure of Arg90Cit BsCM* in its apo form and in complex with mechanistically relevant ligands. In addition to Arg90Cit, this variant contains a second substitution (Asp102Glu) that was introduced for semisynthetic production; this change has no effect on activity (25) but serves to distinguish the BsCM* from the WT BsCM samples. Well-diffracting crystals of WT BsCM and Arg90Cit BsCM* (Fig. S1) were obtained under very similar conditions, and the structures of both apo enzymes were solved to a resolution of 1.6 Å. The models include residues 1–117 of the 127-residue enzyme, with residues 2–115 showing well-defined electron density. R/R_{free} values are 0.14/0.17 for WT BsCM and 0.17/0.22 for apo Arg90Cit BsCM*.

As found previously (18), BsCM adopts a homotrimeric pseudo- α/β -barrel fold with the three active sites positioned at the subunit interfaces (Fig. S2). The overall structures of WT BsCM and Arg90Cit BsCM* are identical within error limits (rms difference of 0.3 Å for C α of residues 2–115, comparing trimers) and, even in the active site, the two structures superimpose well. For Arg90Cit BsCM* we note a somewhat greater conformational freedom for the active-site residues compared with WT BsCM, with residues Arg7, Cys75, Glu78, and Cit90 alternating between two conformations (Fig. 2A). These conformational states are largely coupled through an extensive hydrogen-bonding network, although some variation between the three different active sites of the protein is observed. In all cases, one of the conformational states closely resembles the corresponding conformation in the WT BsCM structure, with ~50% of the residues adopting this conformation. The virtually unchanged affinity for substrate and inhibitor suggest that this increased flexibility cannot account for the dramatically decreased catalytic efficiency of the Arg90Cit variant. Indeed, even more flexible enzymes, such as an engineered monomeric CM from *Methanococcus jannaschii* (30), which exhibits the characteristics of an intrinsically disordered molten globule in the absence of ligand (31, 32), exhibit near native-like activity (30, 33).

The crystal structure of Arg90Cit BsCM* in complex with a transition state analog (compound 4) (34) was solved to 1.8-Å resolution with R/R_{free} values of 0.18/0.23. In this complex, all active-site residues adopt exclusively the WT conformation (Fig. 1C and D), with the exception of Arg63, which is generally disordered. As in the analogous complex with WT BsCM (18), the inhibitor makes numerous contacts with the protein. The side-chain carboxylate of 4 forms a salt bridge with Arg7 and also hydrogen bonds with Cit90 and Tyr108; its hydroxyl group is hydrogen bonded to Glu78 and the backbone amide of Cys75; and its ether oxygen is within hydrogen bonding distance of Cit90. Citrulline differs from arginine in having a neutral urea in place of the cationic guanidinium group (in effect, an NH₂⁺ group is replaced by an oxygen atom); hence the two amino acids are isosteric but differ in net charge. The orientation of the side-chain urea group was assigned based on meaningful hydrogen bonding geometry, which placed the citrulline NH₂ group at a

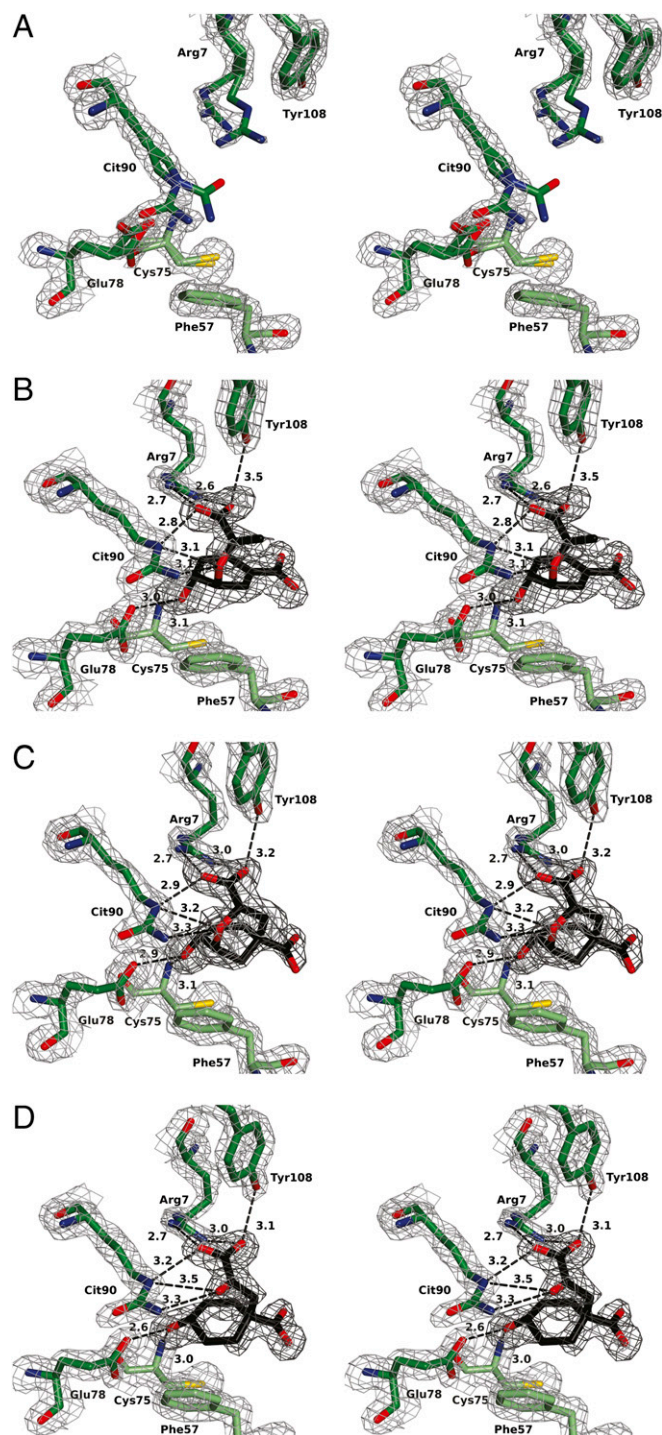


Fig. 2. Snapshots of the BsCM reaction coordinate. Stereo images of the Arg90Cit BsCM* active site (A) without ligand and complexed with (B) chorismate (in the **1b** conformation), (C) transition state analog **4**, and (D) prephenate (**3**). The electron density maps (σ_A -weighted $2F_o - F_c$ maps represented by gray mesh) are contoured at 1.5σ . Contours reflecting the different occupancies of the ligands have been added (dark gray mesh at 0.8σ for **4** and 0.5σ for chorismate/prephenate); real space correlation coefficients for the depicted ligands are 0.86 for chorismate, 0.85 for **4** and 0.88 for prephenate, consistent with well-defined structures. Interactions ($<3.5\text{ Å}$) with the ligands are indicated with broken lines, labeled with the corresponding distances in ångströms. The displayed active sites are between chains E/D, E/D, C/B, and A/C, respectively, which exhibit an rms difference over all atoms of the displayed amino acid side chains of $0.16 \pm 0.03\text{ Å}$.

distance of 3.2 Å from the ether oxygen of **4**. It is evident that Arg90Cit BsCM*, with and without the transition state analog, is very similar to WT BsCM (Fig. 1D). The modest 1.1-kcal/mol decrease in stability of the inhibitor complex compared with that of WT BsCM (**25**) is likely due to the slight disorder observed in the apo Arg90Cit BsCM* binding pocket.

Although the chorismate to prephenate rearrangement is highly exothermic (**35**), the greatly impaired catalytic competence of Arg90Cit BsCM* raised the possibility of structurally characterizing the substrate complex. We therefore soaked the apo crystal in a solution containing 40 mM chorismate. To minimize conversion of chorismate to prephenate due to spontaneous thermal rearrangement or residual enzyme activity, the soaking experiments were performed rapidly (for 5 min) and at reduced temperature (4°C), followed by flash-freezing. The structure was solved to 1.7-Å resolution, with R/R_{free} values of 0.17/0.22. As anticipated, chorismate was not fully converted, allowing the substrate complex to be partially trapped. The relative occupancies of chorismate and prephenate were estimated by comparing the σ_A -weighted $F_o - F_c$ difference maps, refined for different ligand ratios (*Materials and Methods*); final chorismate/prephenate ratios are 0.35/0.25 (chains E/D; Fig. 2B), 0.25/0.3 (chains B/A), 0.3/0.4 (chains C/B), 0.2/0.4 (chains A/C; Fig. 2D) and 0.15/0.3 (chains D/F). The electron density of the ligand mixture is highest for the side-chain carboxylate and the ring hydroxyl group, which have identical positions for substrate and product, whereas the density is lowest for the bonds being broken (C3–O7 in chorismate) or formed (C1–C9 of prephenate). Both ligands adopt conformations similar to the transition state analog **4** (Fig. 2C), with the prephenate complex closely resembling the previously published (**18**) prephenate-bound structure of WT BsCM (Fig. S3). This uniform mode of ligand binding explains why the enzyme's K_d values for binding of prephenate and chorismate are similar (**29**).

The MD simulations performed by Hur and Bruce (**17, 36**) predicted large conformational changes for Glu78, Arg7, Cit90, and Arg116 in the low-activity variant. Specifically, the charge imbalance resulting upon substitution of Arg90 was expected to fundamentally alter the hydrogen-bonding pattern at the active site, causing displacement of the Cit90 side chain and formation of a salt bridge between Arg7 and Glu78 rather than between Arg7 and the side-chain carboxylate of the substrate/transition state analog. As a consequence, the side-chain carboxylate of chorismate was expected to form a salt bridge with Arg116 rather than with Arg7, thereby increasing the distance between the vinyl C9 carbon and C1 of chorismate to $4.5 \pm 0.2\text{ Å}$, compared with $3.5 \pm 0.3\text{ Å}$ calculated for the WT. Nevertheless, as shown in Fig. 1C and D, we find no evidence for this altered binding mode. The crystal structures unequivocally show that the Arg90Cit substitution does not induce a major reorganization of the BsCM active site. The residues of main importance in the catalytic pocket (i.e., Arg7, Phe57, Cys75, Glu78, Cit90, and Tyr108) assume the same conformations as the analogous residues in WT BsCM, with and without bound ligands. Moreover, the C1–C9 distance for chorismate in the Michaelis complex ranges from 3.5 to 3.7 Å in the occupied crystallographically nonequivalent active sites of the variant, matching the calculated WT value very well. This distance is slightly larger than the combined van der Waals radii of the two carbon atoms (1.7 Å each).

Discussion

The structures of the resting Arg90Cit BsCM* enzyme (Fig. 2A) and its complexes with substrate (Fig. 2B), a transition state analog (Fig. 2C), and product (Fig. 2D) provide experimental snapshots of the major states of the CM reaction (see [Movie S1](#) for a morphed animation of the snapshots). The empty enzyme is preorganized for catalysis, binding chorismate in the rare pseudodiaxial conformation primed for pericyclic rearrangement, the

Table 1. Data collection and refinement statistics

Structure	Arg90Cit BsCM*	Arg90Cit BsCM* with 4	Arg90Cit BsCM* with 1 + 3	WT BsCM
X-ray source	MicroMax-007HF	ID23-2, ESRF	ID23-2, ESRF	MicroMax-007HF
Wavelength, Å	1.54187	0.87260	0.87260	1.54187
Space group	<i>P</i> 1	<i>P</i> 1	<i>P</i> 1	<i>P</i> 2 ₁
Unit cell parameters				
<i>a</i> , Å	50.1	50.2	50.2	49.5
<i>b</i> , Å	51.1	50.8	51.3	165.2
<i>c</i> , Å	69.9	69.8	70.1	49.5
α, °	97.8	97.4	97.5	90.0
β, °	93.0	93.2	92.9	118.7
γ, °	101.3	101.0	101.1	90.0
Resolution, Å	43.47–1.61 (1.71–1.61)	49.12–1.80 (1.90–1.80)	49.15–1.70 (1.79–1.70)	43.51–1.59 (1.69–1.59)
<i>I</i> /σ(<i>I</i>)	21.5 (7.2)	23.3 (5.0)	15.5 (4.3)	14.4 (4.4)
No. of reflections				
Observed	135,299 (18,781)	100,173 (11,421)	133,842 (21,371)	227,540 (19,579)
Unique	76,808 (10,808)	53,585 (6,216)	70,311 (11,262)	79,445 (8,930)
Multiplicity	1.8 (1.7)	1.9 (1.8)	1.9 (1.9)	2.9 (2.2)
Completeness, % [†]	88.3 (77.0)	86.1 (61.8)	93.7 (92.8)	85.0 (59.4)
<i>R</i> _{merge} [‡]	0.020 (0.095)	0.021 (0.129)	0.034 (0.173)	0.052 (0.176)
<i>R</i> _{meas} [§]	0.029 (0.135)	0.029 (0.182)	0.049 (0.245)	0.062 (0.222)
<i>R</i> / <i>R</i> _{free} [¶]	0.17/0.22	0.18/0.23	0.17/0.22	0.14/0.17
rms bond length, Å	0.008	0.007	0.007	0.006
rms bond angle, °	1.203	1.051	1.043	1.171
Average <i>B</i> factor, Å ²				
Backbone	16.6	24.7	15.9	12.4
Side chain + water	20.9	31.7	23.3	16.9
All atoms	19.0	28.4	20.1	14.8
No. of atoms				
Protein	5,905	5,604	5,754	5,605
Ligand(s)	n.a.	64	160	n.a.
Solvent	450	356	563	227
Ramachandran, %				
Favored	98.7	99.0	99.0	98.4
Allowed	1.3	1.0	1.0	1.6
Outliers	0.0	0.0	0.0	0.0
PDB ID code	3ZOP	3ZP4	3ZP7	3Z08

Values in parentheses are referring to the highest resolution shell. n.a., not applicable.

[†]The three highest resolution shells are less than 90% complete, except Arg90Cit BsCM* with 1 + 3.

[‡] $R_{\text{merge}} = \sum_h |I_h - \bar{I}_h| / \sum_h \sum_i I_{h,i}$, where $\bar{I}_h = (1/n_h) \sum_i I_{h,i}$.

[§] $R_{\text{meas}} = \sum_h \sqrt{n_h / (n_h - 1)} \sum_i |I_h - \bar{I}_h| / \sum_h \sum_i I_{h,i}$, where $\bar{I}_h = (1/n_h) \sum_i I_{h,i}$ (62).

[¶] $R = \sum ||F_o| - |F_c|| / \sum |F_o|$, where $|F_o|$ and $|F_c|$ are the observed and calculated structure factor amplitudes, respectively. For *R*_{free} the reflections not used in the refinement were used for the calculation.

so-called NAC. Simply constraining the substrate to a reactive conformation is not sufficient for efficient catalysis, however. The kinetic parameters for Arg90Cit BsCM* indicate that populating a NAC-like conformer contributes at most a factor of ~100 toward the 2×10^6 -fold enzymatic rate acceleration over the spontaneous Claisen rearrangement. Because the missing positive charge is the salient feature of the weakly active variant, we conclude that stabilization of the transition state via electrostatic interactions is the main factor determining the catalytic proficiency of BsCM. This conclusion is supported by recent high-level QM/MM studies showing that the higher free-energy barrier for the Arg90Cit mutant is entirely ascribable to less effective transition state stabilization by the neutral citrulline (24, 37). Upon completion of the reaction, the product, which is generated in an unfavorable conformation, is released to bulk solvent.

The finding that electrostatic complementarity in the transition state is a more important contributor to catalysis for BsCM than an active-site geometry tailored to a specific reactive substrate conformer is potentially relevant to understanding enzyme promiscuity and the evolution of enzyme function. Work on arylsulfatase (38), alkaline phosphatase (39), and paraoxonase (40) has revealed that the dominant factor for efficient catalysis is not shape

complementarity to the substrate. More important is the ability of these enzymes to exploit electrostatic preorganization of the active site for the native substrate to stabilize the transition state of promiscuous reactions. Capitalizing on this capacity could be useful for ongoing efforts to design new active sites de novo (4, 5).

Together with previous mutagenesis (26–28) and modeling studies (23, 24, 37), our results provide a unified view of CM catalysis. They highlight the importance of a cationic residue—either an arginine or a lysine—proximal to the ether oxygen of the substrate in the transition state. All known natural CMs, including the structurally unrelated AroQ enzymes from *Escherichia coli* (19), yeast (20), and *Mycobacterium tuberculosis* (41, 42) share this feature, whereas poor catalysts like Arg90Cit or the catalytic antibody 1F7 (43) lack it. Rather than an exception to the classic view of enzymatic catalysis, CM appears to be an archetypical example of the Pauling paradigm.

Materials and Methods

Production of BsCM. WT BsCM was produced in the CM-deficient *E. coli* strain KA13 (44, 45) and purified according to ref. 28. Semisynthetic Arg90Cit BsCM*, which additionally contained the substitution Asp102Glu (*) to prevent a problematic aspartimide formation, was produced according to ref. 25.

This substitution in addition served as a “watermark”, confirming that the Arg90Cit BsCM* variant was indeed present in the crystals. In brief, the WT BsCM fragment (1–87) was produced recombinantly in *E. coli* strain KA13, fused to the *Mycobacterium xenopi* GyrA intein and a chitin-binding domain. It was purified and cleaved on a chitin column to yield the BsCM (1–87) thioester. Arg90Cit BsCM* (88–127) was produced by solid-phase peptide synthesis. Both fragments were coupled by native chemical ligation to yield full-length Arg90Cit BsCM*. The ligated enzyme was purified by ion-exchange chromatography, and its identity was confirmed using mass spectrometry and circular dichroism spectroscopy. The correct oligomerization state was confirmed with analytical size-exclusion chromatography and analytical ultracentrifugation (25). The protein was subsequently concentrated to 3 mM by ultrafiltration for crystallization.

Ligand Production. Chorismate and transition state analog **4** were prepared according to previously described procedures (46, 47). The initial purity of the chorismate stock solution before soaking was estimated to be >90% by monitoring the chorismate concentration by UV absorption measurements at 274 nm during conversion to prephenate using WT BsCM. As chorismate was still converted at a low rate by Arg90Cit BsCM* and also spontaneously rearranges to prephenate in aqueous solutions, the presence of prephenate in the crystal-soaking experiments cannot be avoided.

Crystallization and Crystal Soaking. Crystals of WT BsCM were grown using the hanging drop vapor diffusion technique, by adding 1 μ L well solution, containing 100 mM malic acid, Mes, Tris (MMT) buffer (in molar ratios of 1:2:2, respectively), pH 6.0 (48); 100–150 mM $MgCl_2$; 25% (wt/vol) PEG 1000; and 0.3 mM Na_2S_3 , to 1 μ L concentrated protein solution (3 mM WT BsCM, 10 mM Tris buffer pH 7.5, 2 mM DTT, 0.125 mM EDTA). Large rhomboid-shaped crystals (to approximately 500 μ m in length) grew after 4–5 d at 20 °C. Arg90Cit BsCM* crystals were grown using similar conditions. Here, 0.75 μ L well solution, containing 100 mM MMT buffer pH 6.0, 50–150 mM $CaCl_2$, 24–26% (wt/vol) PEG 1000, and 0.3 mM Na_2S_3 , was added to 0.75 μ L concentrated protein solution (3 mM Arg90Cit BsCM*, 20 mM Tris buffer pH 8.0, 0.6 mM PMSF, 0.3 mM Na_2S_3). Thin, elongated rhomboid-shaped crystals (to approximately 500 μ m in length) grew after 3–4 d at 20 °C. The crystals that were not used for soaking experiments were briefly dipped into a cryoprotectant solution [freshly prepared reservoir solution, with 5% (vol/vol) PEG 400 added] before flash-freezing in a stream of dry nitrogen gas at 100 K or liquid nitrogen. For soaking, reservoir solution containing in addition 5% (vol/vol) PEG 400 and a lower MMT buffer concentration of 27 mM was freshly prepared. Additionally, the solution contained 10 mM transition state analog **4** or 40 mM chorismate, respectively, but lacked $CaCl_2$ for better soaking efficiency. Estimated twice the volume of the solution was then pipetted into the drop containing the crystals. Soaking of **4** into the crystal was performed at room temperature for 1.5 h before freezing; chorismate soaking was performed at 4 °C for 5 min.

X-Ray Diffraction Data Collection. For WT BsCM and Arg90Cit BsCM* (apo) crystals, the X-ray diffraction experiments were performed using a Rigaku MicroMax-007HF instrument. The data were recorded on an R-Axis-IV++ detector at a temperature of 100 K. For Arg90Cit BsCM* crystals soaked with chorismate or **4**, data were collected at the European Synchrotron Radiation Facility (ESRF) in Grenoble, France (Beamline ID23-2) at a wavelength of 0.8726 Å and a temperature of 100 K; the data were recorded on a Mar/ Rayonix 3 $\times$ 3 Mosaic 225 detector. Further information and data collection statistics are listed in Table 1.

Data Processing, Structure Determination, and Refinement. The raw diffraction images were processed using XDS (49). Scaling of the data were performed using SCALA (50) and the structures were determined by molecular replacement using the program Phaser (51), and programs of the CCP4 Suite (52). The WT BsCM crystal structure (residues 1–117) with the Protein Data Bank (PDB) ID code 1DBF (53) was used as a search model. The final models were obtained by iterating cycles of refinement and model building using REFMAC5 (54) and Coot (55), respectively (for refinement statistics, see Table 1). For cross-validation, 5% of the reflections were selected at random. In the case of WT BsCM, the data were almost perfectly twinned, resulting in the apparent space group C22₁, rather than P2₁. A perfectly twinned dataset was generated by averaging the twin-related structure factor amplitudes, and intensity-based twin refinement was then used for REFMAC5 runs. In the twinned case, the test set was selected as 20 thin shells over the entire resolution range, ensuring that twin-related reflections were always in the same set. Anisotropic *B* factor refinement was used for the apo structures; for the structures of Arg90Cit BsCM* bound to **4** and chorismate/prephenate, isotropic *B* factor refinement was used, but included translation/libration/screw-motion (TLS) anisotropy refinement in the final runs with 18–20 TLS groups per protein chain. TLS groups were defined using the TLSMD web server (56, 57). TLS refinement runs were performed in the PHENIX suite (58). In all cases, only parameters belonging to atoms of the protein model were refined for the first few cycles, and then most ordered water molecules were added in the course of two cycles at positions where there were sufficient electron density and sensible hydrogen-bonding partners. In the case of **4**- and chorismate/prephenate-bound structures, no water molecules or ligands were modeled in close proximity of the enzymatic pockets during these initial cycles. Subsequently, the ligands were modeled into the active sites, if at least part of the ligand electron density was clearly visible at 1.0 σ (σ_A -weighted $2F_o - F_c$ map). During the refinement process, the phases improved significantly, allowing ligand modeling in additional active sites. The occupancies of the ligands were adjusted manually, by matching the *B* factors of atoms in hydrogen bonds with the enzyme. For the structures with chorismate/prephenate bound in the active site, models of both ligands were added to the active sites into the same electron density, initially with equal occupancies, to represent the mixture present in the crystal. The final relative occupancies for the ligands were then estimated by comparing difference maps refined for different ligand ratios. The electron density of citrulline is indistinguishable from that of an arginine residue, unless atomic resolution data are available (1.0 Å or better). Here, the watermark substitution Asp102Glu served its purpose, confirming the proper identity of Arg90Cit BsCM*. For ligand-bound structures the head group orientation was determined based on chemical sense, relying on the fact that there was only one potential bonding partner with sensible H-bonding geometry close to the citrulline amide group, thereby determining its orientation. In this orientation, Cit90 is positioned within 3.2 Å (average over four active sites) from the ether oxygen of **4** (Fig. 1 *B* and *C*) and within 3.0 Å from the corresponding O7 of chorismate. Structure validation was performed using Molprobity (59) and its implementations in Coot or PHENIX, structure superimpositions were performed with the CCP4-supported program Superpose (60), and images of the models were prepared with PyMOL (61).

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